

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EET307**Course Name: SYNCHRONOUS AND INDUCTION MACHINES**

(GRAPH SHEET NEEDED)

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|---|---|
| 1 | Discuss about the different types of windings in alternator | 3 |
| 2 | What are the causes of harmonics in the induced voltage of an alternator and how its effects are minimised? | 3 |
| 3 | Draw and explain the phasor diagram of a salient pole alternator supplying to a lagging power factor load | 3 |
| 4 | Enumerate the requirement of proper parallel operation of alternators | 3 |
| 5 | Discuss about any two starting methods of synchronous motors | 3 |
| 6 | Derive expressions for starting torque in three phase induction motor | 3 |
| 7 | Explain about working of star delta starter for three phase induction Motor | 3 |
| 8 | Explain about V/f speed control method in three phase induction motors | 3 |
| 9 | Compare the operation of induction Generator in grid connected mode and self excited mode | 3 |
| 10 | Explain about working of split phase induction motor | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|--|---|
| 11 | a) List the advantages of using stationary armature in alternator | 7 |
| | b) A 3 phase alternator, 50 Hz, 8 pole, has 180 conductors per phase, flux/pole is .054wb, find the induced line emf (assume $k_p = k_d = 1$) | 7 |
| 12 | a) What is the significance of winding factor in the induced voltage of induction motor? Derive an expression for the same | 7 |
| | b) Explain different types of alternators based on rotor construction | 7 |

Module -2

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|----|--|----|
| 13 | a) Find the full load regulation by MMF method of a 100 kVA, 2kV 3 phase 50 Hz star connected alternator at 0.8 power factor lag having the following test data, armature resistance = 0.2Ω per phase | 14 |
|----|--|----|

If in A	10	20	25	30	40	50
Vt (line voltage)	800	1500	1760	2000	2350	2600
Isc in A	-	20	25	30	-	-

- 14 a) With circuit diagram explain about synchronisation of alternator using bright lamp method 7
- b) Explain slip test and calculation of X_d and X_q 7

Module -3

- 15 a) A 75KW,400V,4 pole ,3 phase star connected synchronous motor has (0.04+j0.4)/phase ,compute for full load 0.8pf lead, armature current and gross mechanical power developed assuming an efficiency of 92.5% 7
- b) Draw & explain about the power- angle characteristics of synchronous motor 7
- 16 a) Discuss power stage diagram of induction motor 4
- b) The power input of a 500V,50Hz, 6 pole, 3 phase induction motor running at 975 rpm is 40KW,the stator losses is 1 KW and friction and windage losses are 2KW.find slip,rotor copper losses,BHP and efficiency 10

Module -4

- 17 a) Draw the circle diagram of a 20 HP,400V,50Hz,3 phase star connected induction motor from the following test data (line values) 14
- No load test : 400V 9A pf 0.2
- Blocked rotor test : 200V 50A pf 0.4
- from circle diagram find line current and power factor at full load and maximum power output
- 18 a) Explain about different braking methods of induction motor 7
- b) Explain how high starting torque and good operating efficiency is achieved in double cage induction motor 7

Module -5

- 19 a) Explain why single phase induction motor is not self starting 7
- b) Draw and discuss about speed - torque characteristics of induction generator 7
- 20 a) Explain about induction Generator 7
- b) Explain capacitor start capacitor run single phase induction motors 7